Flow Matching Policy Optimization with Mirror Descent and Entropy Constraints

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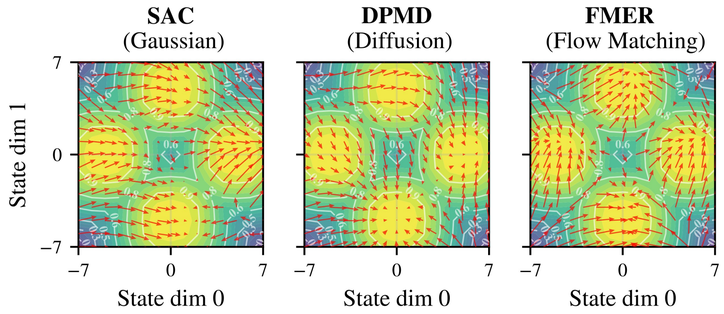
TL;DR Diffusion (SDE) policies have intractable entropy and unstable gradients. **Flow matching (ODE) could fix both.** FMER turns policy mirror descent into a **weighted flow-matching regression** and adds an **exact, tanh-corrected entropy constraint**, giving the best results on sparse-reward FrankaKitchen.

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What is the problem?

Good behaviour is often multimodal

Gaussian policies pick one of several equally good actions.



2D multi-goal task (bright = goals, arrows = chosen actions). SAC and DPMD drift to favourite goals; FMER heads for the nearest of four equally good goals.

But diffusion (SDE) policies can hit two walls

✗ **Entropy is intractable**
→ exploration relies on heuristics

✗ **Gradients through the denoising chain**
→ expensive, high-variance, unstable

Key insight: switch to an ODE flow-matching policy

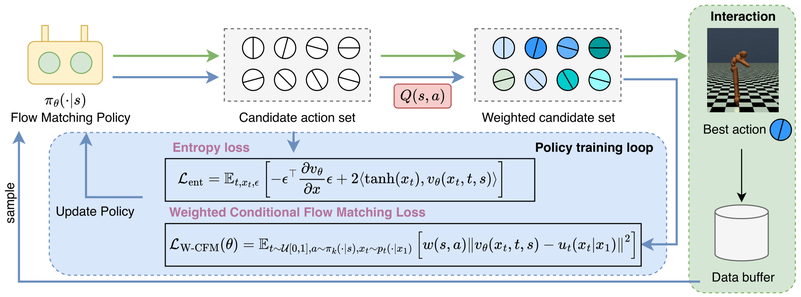
Entropy becomes the divergence of the velocity field, and training becomes plain regression: **no simulation, no backprop through the ODE**.

	Mirror descent	Entropy constraint	Unbiased entropy	Simulation-free
DIPO	×	×	×	✓
DPMD	✓	×	×	✓
QVPO	×	✓	×	✓
SAC-Flow	×	✓	×	×
FLAME	×	✓	×	×
FMER (ours)	✓	✓	✓	✓

FMER: one simulation-free update

A **policy mirror descent** step under a **minimum-entropy constraint**:

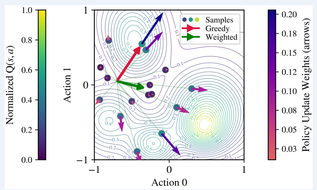
$$\mathcal{L}_{\mathrm{FMER}}(\theta) = \mathcal{L}_{\mathrm{W-CFM}}(\theta) + \alpha \mathcal{L}_{\mathrm{ent}}(\theta)$$



Mirror descent = weighted flow matching

Theorem 1. Minimisers of the weighted CFM loss = mirror-descent target:

$$\mathcal{L}_{\mathrm{W-CFM}} = \mathbb{E}_{t,s,a,x_0}\left[w(s,a)\,\|v_\theta(x_t,t,s) - (x_1 - x_0)\|^2\right]$$



$$\Longrightarrow \quad \pi_{\theta^*}(a|s) = \frac{\pi_k(a|s)\exp(A(s,a)/\tau)}{Z(s)}$$

- In practice: sample $M = 8$ candidates from π_k; weight each by softmax of $A(s,a)/\tau$.
- Pure regression: **no backprop through the ODE**
- Soft weights pool all good candidates; greedy top-1 (red) gets stuck

Analytical entropy in a bounded action space

Entropy = integrated divergence of the velocity field; for bounded actions a = tanh(x), **Prop. 1** adds the exact **tanh correction**:

$$\mathcal{L}_{\mathrm{ent}} = -\,\mathbb{E}_{t,x_t}\left[\nabla_x \cdot v_\theta \;-\; 2\left\langle \tanh(x_t),\, v_\theta \right\rangle\right]$$

Self-tuning trust region

$$\mathrm{ESS}(s) = \frac{1}{\sum_l \hat{w}_l(s)^2} \in [1, M]$$

τ sets the trust-region size; FMER adapts τ to hold Efficient Sample Size (ESS) at a target, so no per-task tuning.

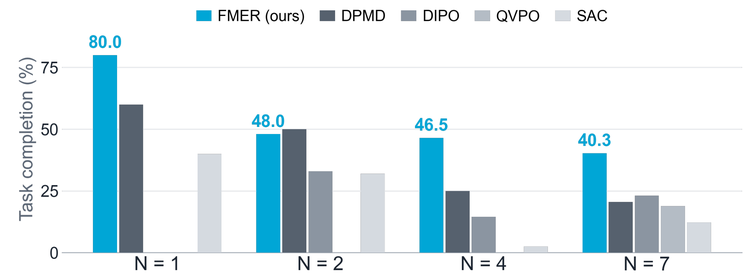
Does it work?

1.7× best baseline, FrankaKitchen N = 7
40.3% vs 23.1% (DIPO)

1.9× best baseline, FrankaKitchen N = 4
46.5% vs 25.0% (DPMD)

Sparse-reward FrankaKitchen

Reward only on sub-task completion · 280 steps · 1M steps · 5 seeds



Not shown: FLAME, TD3, PPO, SAC-Flow (all ≤ 18.3%).

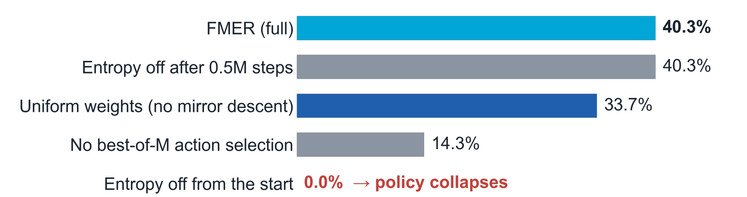
FMER completes **5 tasks** in 280 steps (best baseline DIPO: 3)



Dense-reward MuJoCo: still competitive

HalfCheetah	Walker2d	Humanoid	Hopper
12.3k	**5.28k**	**5.29k**	**2.91k**
#1 of 9	#1 of 9	#2 of 9	#7 of 9

Every component matters (FrankaKitchen N = 7)



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